INCH-POUND

MIL-P-20449C (AR)
AMENDMENT 3
9 February 1999
SUPERSEDING
AMENDMENT 2
1 April 1983

MILITARY SPECIFICATION

PRIMER, PRECUSSION, M61 PARTS AND LOADING, ASSEMBLING AND PACKING

MIL-P-20449C was inactivated after 10 July 1996 for new design.

This Amendment forms a part of Military Specification MIL-P-20449C (AR) dated 18 October 1968, and is approved for use by the US Army Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete the following documents:

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 – Inspection Equipment, Supply and Maintenance of
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 – Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Add the following Document:

* "MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product"

DRAWINGS: Add the following:

"8815988 - Primer Holder
8815993 - Pin, Firing
8816003 - Primer, Holder Adapter
8816015 - Ball, Steel"

AMSC N/A FSC 1390
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PAGE 2

Delete: "Publications US ARMY MUNITIONS COMMAND ET-8831157 – Equipment Tabulation".

PAGE 3

* 4.1.1 Responsibility for inspection. Delete references to MIL-STD-109 and MIL-A-2550.

PAGE 4

4.2.1 Delete in its entirety and substitute the following:

"4.2.1 Submission. The contractor shall submit a first article sample (see 6.1c) as designated by the Contracting Officer for evaluation. The first article sample shall consist of the following:

a. 25 each of every component part, component subassembly, and final assembly for examination (see 4.2.2).

b. 450 each final assemblies for sensitivity testing (see 4.3.3.1.1).

c. 300 each final assemblies for dud testing (see 4.3.3.1.2).

All parts, subassemblies and final assemblies shall have been produced by the contractor or furnished by a supplier using the same production processes, procedures and equipment which be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms the requirements of the contract and shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials (see 4.1.1e). A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

PAGE 5

4.2.2 Delete in its entirety and substitute the following:

"4.2.2 Inspections to be performed. Component parts, component subassemblies and final assemblies will be subjected by the Government to any or all of the examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings."

* 4.3.1 Lot formation. Delete "MIL-STD-105" and substitute "MIL-STD-1916".

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* 4.3.2 Delete in its entirety and substitute the following"

"4.3.2 Classification of characteristics. There are three classes of characteristics covered in this document. These are critical characteristics, major characteristics and minor characteristics. (See MIL-STD-1916 for definitions of critical, major and minor classification of characteristics.). Unless otherwise specified herein or provided for in the contract, alternative conformance procedures, methods, or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation. When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified conformance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable."

* Delete paragraphs 4.3.2.1 through 4.3.2.11 in their entirety and substitute the following:

CONFORMANCE INSPECTION
"CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8831158	
4.3.2.1	Cup, Primer			NEXT HIGHER ASSEMBLY 8831157	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE	Code No.
Critical	None defined				
Major					
101	Outside diameter, min	Level IV	3.1	Gage	01001
102	Inside diameter, max	Level IV	3.1	Gage	01002
103	Thickness through Bottom	Level IV	3.1	Gage	01003
104	Total length	Level IV	3.1	Gage	01004
Minor					
201	Evidence of poor Workmanship	Level II	3.4	Visual	01005
NOTES:					

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CLASSIFICATIOIN OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8831159	
4.3.2.2	Anvil			NEXT HIGHER ASSEMBLY 8831157	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE	Code No.
Critical	None defined				
Major					
101	Profile of point	Level IV	3.1	Gage	02001
102	Diameter, min	Level IV	3.1	Gage	02002
103	Thickness, min	Level IV	3.1	Gage	02003
104	Split	Level IV	3.1	Visual	02004
Minor					
201	Length between grooves	Level II	3.1	Gage	02005
202	Evidence of poor workmanship	Level II	3.4	Visual	02006
NOTES:					

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8831157	
4.3.2.3	Primer, prior to assembling cover			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE	Code No.
Critical	None defined				
Major					
101	Complete charge, or portion of charge missing	Level IV	3.2	Visual	03001
Minor					
201	Evidence of poor workmanship	Level II	3.4	Visual	03002
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8831157	
4.3.2.4	Primer, prior to assembly of anvil			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE	Code No.
Critical	None defined				
Major					
101	More than one cover	Level IV	3.2	Visual	04001
Minor					
201	Cover missing, torn, cut or improperly inserted	Level II	3.2	Visual	04002
202	Evidence of poor workmanship	Level II	3.4	Visual	04003
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8831157	
4.3.2.5	Primer assembly			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical					
1	Cup split or cracked	100 %	3.2	Visual	05001
Major					
101	Total length, max	Level IV	3.2	Gage	05002
102	Outside diameter, max	Level IV	3.2	Gage	05003
103	Anvil missing, inverted, cocked, loose or damaged to extent that function is impaired	Level IV	3.2	Visual	05004
104	Explosive on exterior	Level IV	3.2	Visual	05005
105	Cup dented or distorted	Level IV	3.2	Visual	05006
Minor					
201	Evidence of poor Workmanship other than specified in Critical above	Level II	3.4	Visual	05007
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861059	
4.3.2.6	Inner set-up box, prior to sealing			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical	None defined				
Major					
101	Tray damaged to extent that items touch each other	Level IV	5.1	Visual	06001
Minor					
201	Primer missing	Level II	5.1	Visual	06002
NOTES:					

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861059	
4.3.2.7	Inner set-up box			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical	None defined				
Major	None defined				
Minor					
201	Box damaged to extent that contents are exposed	Level II	5.1	Visual	07001
202	Label missing, mislocated or incompletely sealed	Level II	5.3	Visual	07002
	Marking incorrect or illegible	Level II	5.3	Visual	07003
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861059	
4.3.2.8	Outer set-up box			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE	
Critical	None defined				
Major					
101	Sealing strip missing, mislocated or incompletely sealed	Level IV	5.1	Visual	08001
Minor					
201	Box damaged to the extent that contents are exposed or liable	Level II	5.1	Visual	08002
202	to become exposed	Level II	5.3	Visual	08003
203	Label missing, mislocated or incompletely sealed	Level II	5.3	Visual	08004
	Marking incorrect or illegible				
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861059	
4.3.2.9	Outer set-up box sealed in barrier bag			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical	None defined				
Major					
101	Bag cut or torn	Level IV	5.1	Visual	09001
	Seal incomplete or mislocated	Level IV	5.1	Visual	09002
Minor					
201	Marking missing, incorrect or illegible	Level II	5.3	Visual	09003
NOTES:					

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861060	
4.3.2.10	Packing box, prior to closing			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical	None defined				
Major					
101	Carton missing	Level IV	5.1	Visual	10001
Minor					
201	Carton inverted	Level II	5.1	Visual	10002
202	Filler missing	Level II	5.1	Visual	10003
NOTES:					

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER 8861060	
4.3.2.11	Sealed packing box			NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**	**Code No.**
Critical	None defined				
Major					
101	Box damaged to extent that contents are exposed	Level IV	5.1	Visual	11001
102	Strapping missing, broken or loose	Level IV	5.1	Visual	11002
Minor					
201	Marking missing or incorrect	Level II	5.3	Visual	11003
202	Strapping improperly assembled	Level II	5.1	Visual	11004
203	Contents loose	Level II	5.1	Visual	11005
204	Evidence of poor workmanship	Level II	3.4	Visual	11006
NOTES:					

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4.3.4 Delete in its entirety and substitute the following:

"4.3.4 Inspection equipment. Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (see 4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection designs and approval."

4.3.4.1, 4.3.4.2 Delete in their entirety.

4.4.1.1 Add after the last sentence: "Method of inspection shall be in accordance with drawing 8815988, 8815993, 8816003 and 8816015."

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6.3 Delete in its entirety and substitute the following:

"6.3 Inspection equipment designs. Inspection equipment designs are of two types; Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in section 4. Designs responsibilities for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Delete in its entirety and substitute the following:

"6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein."

6.3.2 Delete and substitute the following:

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"6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which

provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.2.c). Designs shall be of the category and form (Per MIL-D-1000) specified in the Contract Data Requirements List (DD From 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Designs" for guidance. The specification number and the applicable 5-digit defect code number (or other specific identifying information) from section 4 of this specification shall be referenced on each contractor design, together with the component of assembly drawing number and revision letter to which the specified design applies."

6.3.3 Delete and substitute the following:

"6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabrication or procuring equipment. Designs shall be submitted for approval to TACOM-ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000 in accordance with stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submissions of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. TACOM-ARDEC design review will normally be accomplished within one month after receipt.

6.3.4 Delete in its entirety

6.3.5 Delete in its entirety.

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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